

Project-Nira: Democratising Environmental Data

Open-source hardware for real-time, flow-through micro-plastic detection



An invisible crisis requires a visible response



Micro-plastics are now ubiquitous in rivers, lakes, and groundwater. But if we cannot measure the problem locally, we cannot respond to it effectively.

The barrier to entry is ₹10+ Lakh



Requires advanced spectroscopy

Requires extensive lab sample preparation

Requires trained specialists

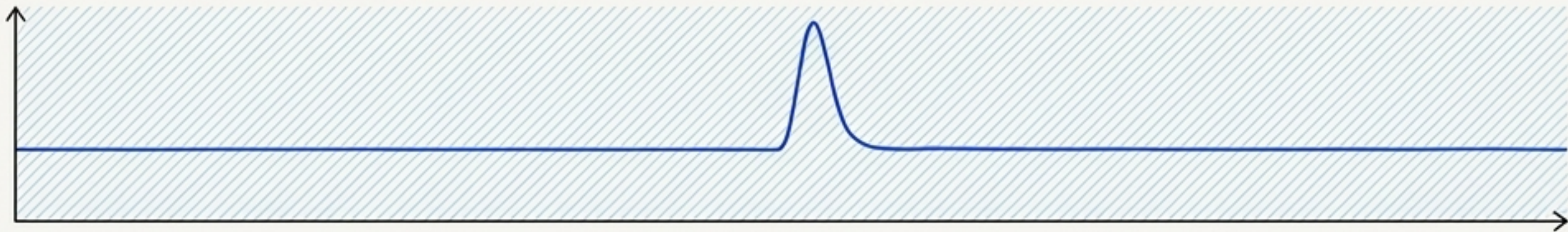
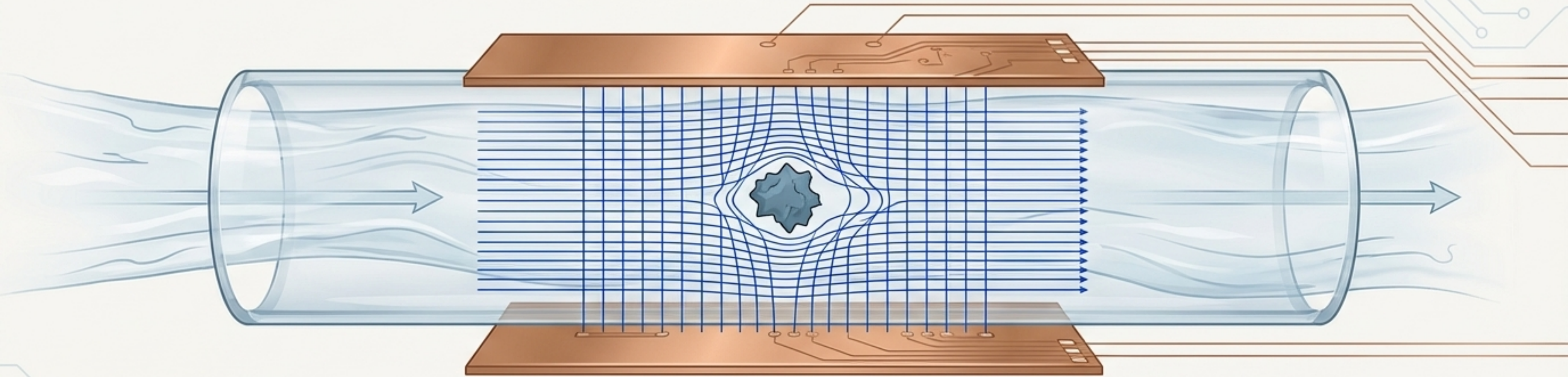
Current environmental monitoring is lab-bound and proprietary.
Local communities are entirely priced out of understanding their own water.



The Accessibility Matrix

Dimension	Traditional Lab Detection	Project-Nira
Method	Advanced Spectroscopy	Impedance-Based Flow-Through
Hardware Cost	₹10,000,000+	Affordable Off-the-Shelf Components
Environment	Sterile, Lab-Bound	Rugged, Field-Deployable
Operator	Trained Specialists	Citizen Scientists & Students
Intellectual Property	Closed, Proprietary	Fully Open-Source & Modular

The Physics of the Flow-Through Method



Changes in electrical impedance (differential capacitance) instantly indicate the presence of suspended particles—no chemistry degree required.

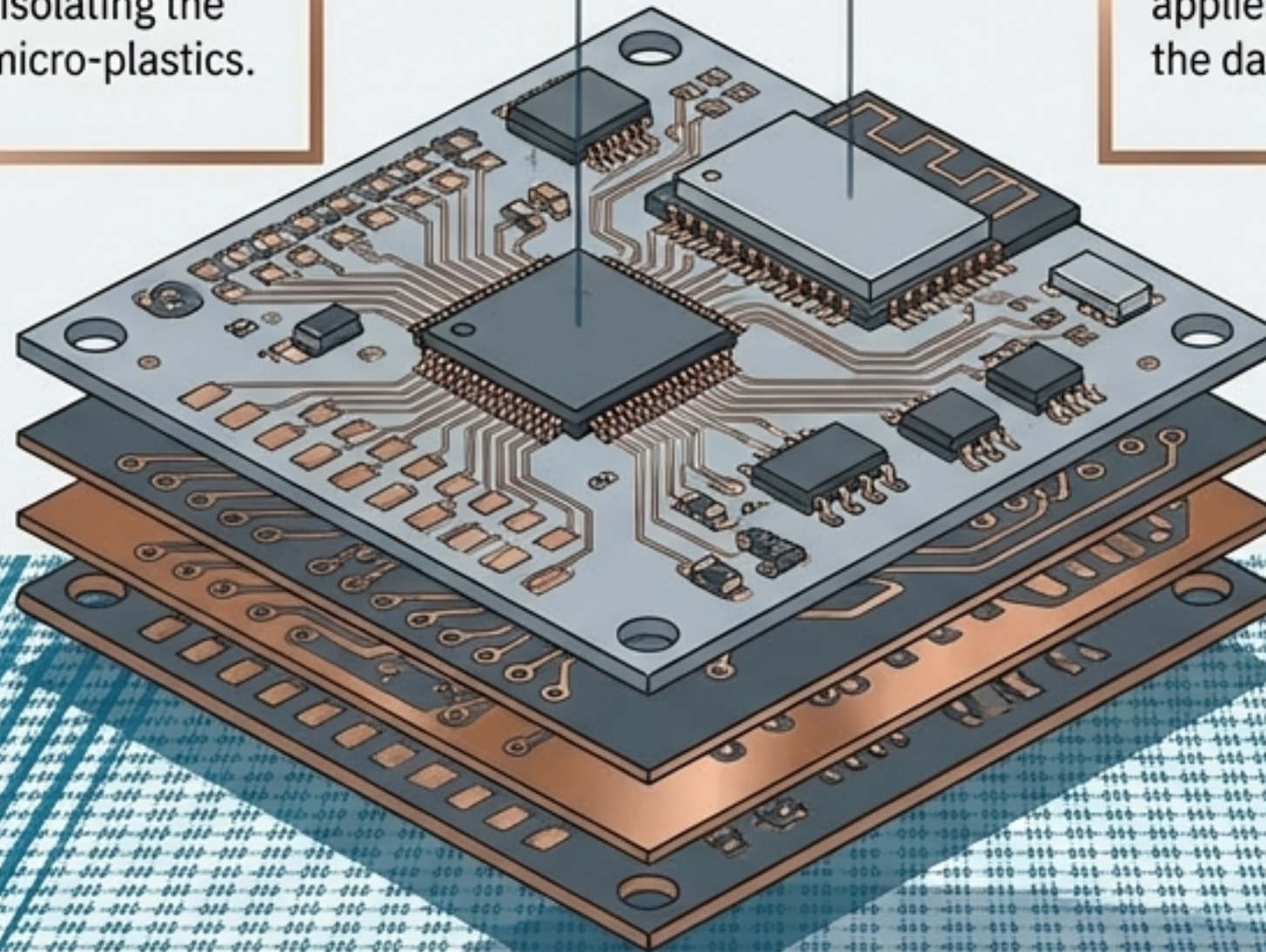
The Open-Hardware Edge Stack

Texas Instruments FDC1004

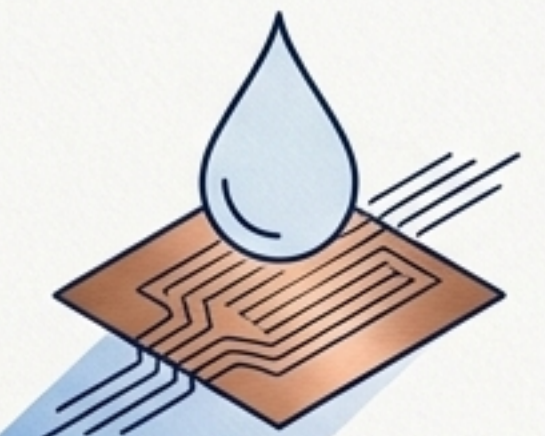
High-noise-immunity capacitive-to-digital converter. Handles the raw analogue differential capacitance readn reading from the sensor plates, isolating the minute signal fluctuations caused by micro-plastics.

ESP32 Microcontroller

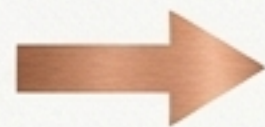
Low-cost, highly modular edge compute. Processes the digital signal from the FDC1004, applies noise-filtering algorithms, and prepares the data packets for wireless transmission.



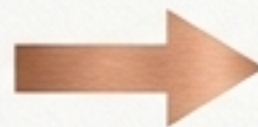
The Edge-to-Cloud Data Pipeline



**Analogue
Impedance Shift**



**Digital Conversion
& Packetization**



**Wireless
Transmission**



**InfluxDB
Time-Series Bucket**

InfluxDB is optimized specifically for high-frequency, time-stamped sensor data, making it the perfect repository for real-time flow-through metrics.

Engineering for Empowerment

**Low-Cost
Off-the-Shelf Parts**



Enables Citizen Scientists
(Schools and local communities
can build their own units)

**Portable &
Battery-Powered**



Allows Rural Monitoring
(Testing happens at the water source,
not hours away in a city lab)

Fully Open-Source IP



Global Reproducibility
(Anyone can modify, repair, or improve
the system without vendor lock-in)

**Environmental monitoring tools must
be accessible, not proprietary.**

A Decentralised Global Water Quality Grid



Project-Nira is not just a sensor. It is the foundational hardware node for a community-owned, open-source map of global water health. Micro-plastic pollution is a global issue; the solution must be locally accessible.